

Properties of a function

Zeros, sign, monotonicity, parity and boundedness — the five questions asked of every function for the rest of the course.

LESSON 19–21

3 × 40 MIN

ALGEBRA 10, §1.6

P1 · 2.2, 3.1

LESSON CLOCK

12

20'

20'

14'

55'

14'

Zeros and the sign chart	12 min
Increasing and decreasing	20 min
Even and odd	20 min
Boundedness	14 min
Practice across all five	55 min
Homework	14 min

LEARNING OBJECTIVES

- Find the zeros of a function and the intervals where it is positive or negative.
- Decide whether a function is increasing or decreasing on an interval.
- Test a function for even and odd symmetry algebraically.
- Say whether a function is bounded above, below, or neither.

TEXTBOOK

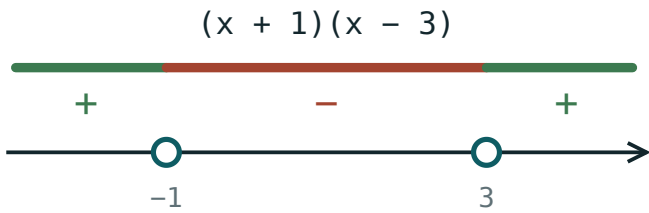
National	pp. 59–70
Cambridge	Pure Mathematics 1, pp. 30–36, 58–62
Workbook	P1 Exercise 2B, 3A

01

Explanation

Zeros and the sign chart

A **zero** of f is an input with $f(x) = 0$ — where the graph meets the x -axis. Between consecutive zeros a continuous function cannot change sign, so the zeros cut the domain into intervals on which the sign is constant.



The zeros split the line into three intervals. Test one point in each and the whole interval follows.

THE METHOD, IN FOUR STEPS

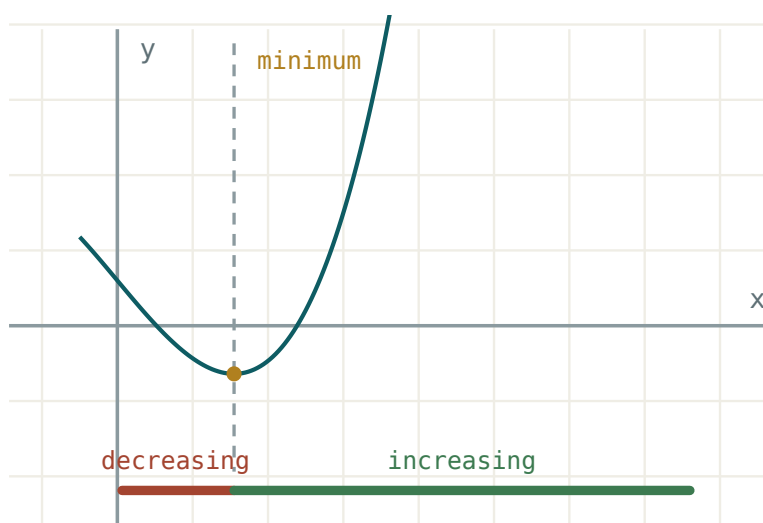
Find the zeros · mark them on a number line, together with any excluded points · test one convenient point in each interval · write the answer as intervals.

For $f(x) = x^2 - x - 6 = (x - 3)(x + 2)$ the zeros are -2 and 3 . Testing $x = 0$ gives -6 , so $f < 0$ on $(-2, 3)$ and $f > 0$ outside it.

Increasing and decreasing

THE DEFINITIONS

f is **increasing** on an interval when $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$ for every pair in it, and **decreasing** when $x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$. A function that is one or the other on the whole domain is **monotonic**.



Decreasing to the left of the vertex, increasing to the right. Neither word describes the whole domain.

ALWAYS NAME THE INTERVAL

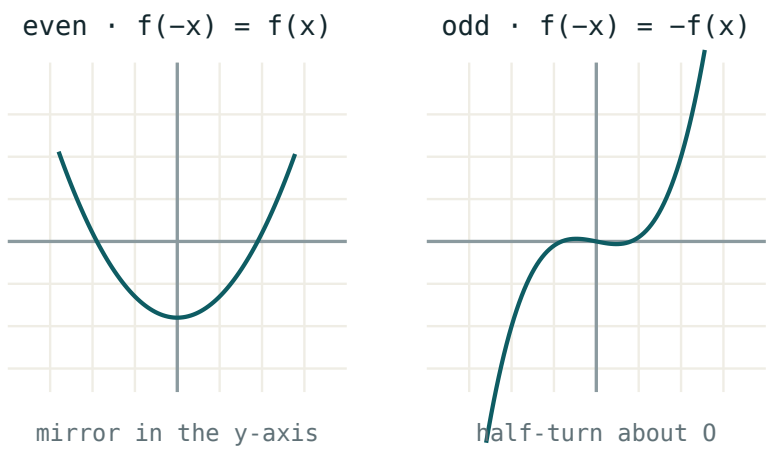
" $y = x^2$ is increasing" is false. " $y = x^2$ is increasing on $x \geq 0$ " is true. Monotonicity is a property of a function **on an interval**, never of a function alone.

A monotonic function is automatically one-to-one, and therefore has an inverse — which is why this property matters so much in the next chapter.

Even and odd

Two symmetries are worth naming, because they halve the work of sketching:

$$\text{even: } f(-x) = f(x) \quad \text{odd: } f(-x) = -f(x)$$



An even graph folds onto itself in the y-axis. An odd graph maps onto itself under a half-turn about the origin.

FUNCTION	$F(-X)$	VERDICT
x^2	x^2	even
x^3	$-x^3$	odd
$x^2 + x$	$x^2 - x$	neither
$ x $	$ x $	even
$\frac{1}{x}$	$-\frac{1}{x}$	odd

“NEITHER” IS A REAL ANSWER

Most functions are neither even nor odd. The only function that is both is $f(x) = 0$. Also: the domain must itself be symmetric about 0, or the question cannot even be asked.

Boundedness

f is **bounded above** if there is a number M with $f(x) \leq M$ for every x in the domain, and **bounded below** if some m has $f(x) \geq m$. Bounded on both sides is simply **bounded**.

FUNCTION	BELOW	ABOVE
x^2	yes, by 0	no
$-x^2 + 5$	no	yes, by 5
$\sin x$	yes, by -1	yes, by 1
$2x + 1$	no	no
$\frac{1}{1 + x^2}$	yes, by 0	yes, by 1

Boundedness is exactly a statement about the **range**: bounded above means the range has a ceiling. So finding the range answers this property for free.

02

Worked examples

EXAMPLE 1

Find the zeros and the sign intervals of $f(x) = x^3 - 4x$.

1

$x(x^2 - 4) = x(x - 2)(x + 2)$
Factorise fully.

2

Zeros $-2, 0, 2$.
Four intervals.

3

Test $-3, -1, 1, 3$: $-15, 3, -3, 15$.

4

Signs alternate.

ANSWER

$f > 0$ on $(-2, 0) \cup (2, \infty)$; $f < 0$ on $(-\infty, -2) \cup (0, 2)$

EXAMPLE 2 Decide whether $f(x) = x^3 - 3x$ is even, odd or neither.

① $f(-x) = (-x)^3 - 3(-x)$
Substitute, do not guess.

② $= -x^3 + 3x$

③ $= -(x^3 - 3x) = -f(x)$

ANSWER

Odd

EXAMPLE 3 On which intervals is $f(x) = x^2 - 6x + 5$ decreasing, and is it bounded?

① Complete the square: $(x - 3)^2 - 4$.
Vertex $(3, -4)$.

② A positive parabola falls to the left of its vertex.
Decreasing on $x \leq 3$.

③ Range $y \geq -4$.
Bounded below only.

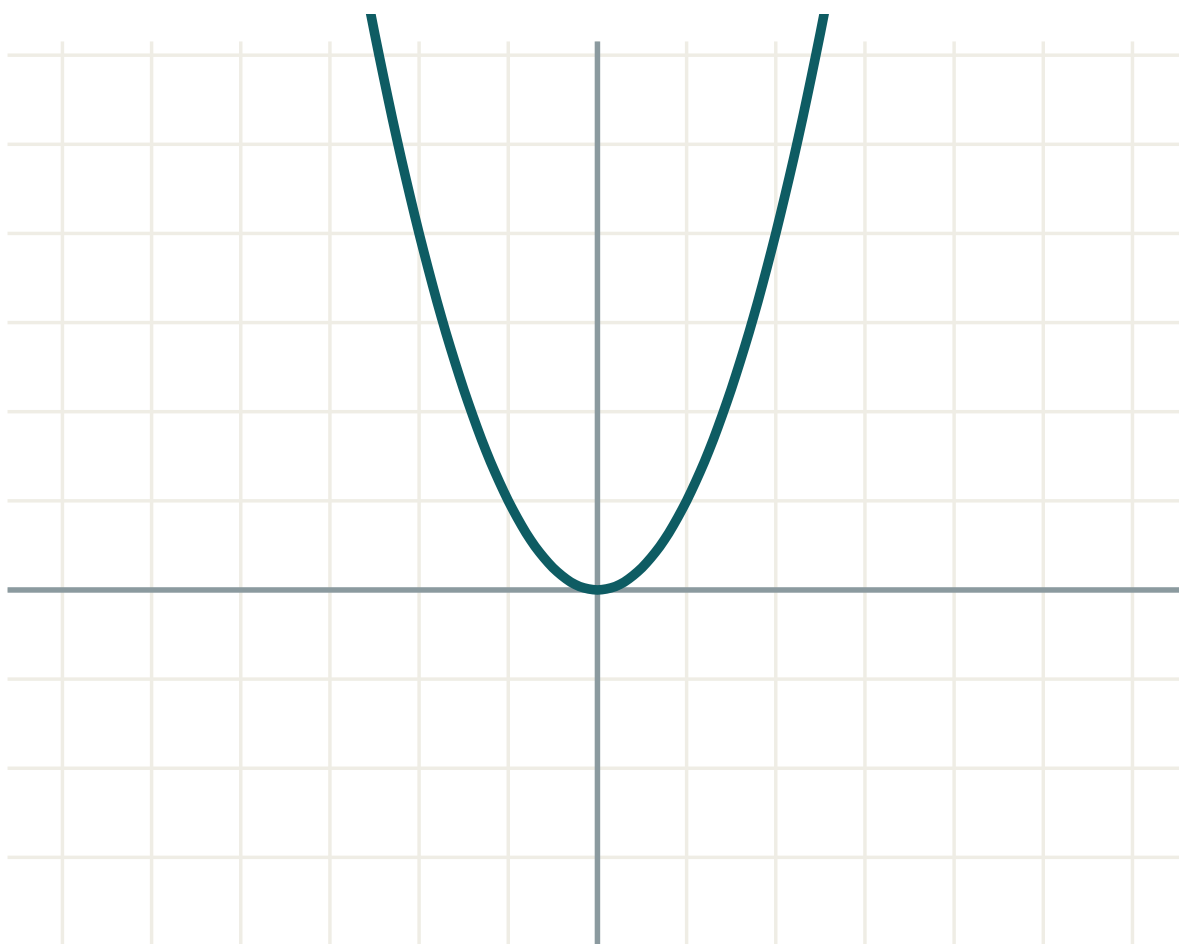
ANSWER

Decreasing on $(-\infty, 3]$; bounded below by -4 , not above

03 Interactive model

Move the point along the curve and watch the sign, the direction and the symmetry all reported at once.

Five properties on one curve



a 1

b 0

c 0

equation $y = f(x)$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Zero of a function	Funksiya nuli	Ноль функции
Sign of a function	Funksiya ishorasi	Знак функции
Increasing function	O'suvchi funksiya	Возрастающая функция
Decreasing function	Kamayuvchi funksiya	Убывающая функция
Monotonic	Monoton	Монотонная
Even function	Juft funksiya	Чётная функция
Odd function	Toq funksiya	Нечётная функция
Bounded above	Yuqoridan chegaralangan	Ограничена сверху
Bounded below	Quyidan chegaralangan	Ограничена снизу
Interval of constant sign	Ishora saqlanish oralig'i	Промежуток знакопостоянства

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$f(-x) = f(x)$ means the function is:

$y = x^2$ is increasing on:

The zeros of $f(x) = (x - 1)(x + 4)$ are:

$f(x) = \sin x$ is:

bounded

bounded below only

bounded above only

unbounded

A monotonic function is always:

even

periodic

one-to-one

bounded

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Zeros of $f(x) = x - 5$

ANSWER $x = 5$

2. Zeros of $f(x) = x^2 - 16$

ANSWER $x = \pm 4$

3. Is x^4 even or odd?

ANSWER even

4. Is x^5 even or odd?

ANSWER odd

5. Is $y = 3x + 1$ increasing or decreasing?

ANSWER increasing

6. Is $y = 7 - 2x$ increasing or decreasing?

ANSWER decreasing

7. Is $y = x^2$ bounded below?

ANSWER yes, by 0

Medium · the standard question

7 PROBLEMS

1. Sign intervals of $(x - 1)(x - 4)$

ANSWER > 0 outside $[1, 4]$, < 0 inside

2. Is $f(x) = x^2 + x^4$ even, odd or neither?

ANSWER even

3. Is $f(x) = x^3 + x$ even, odd or neither?

ANSWER odd

4. Is $f(x) = x^2 + x$ even, odd or neither?

ANSWER neither

5. Where is $y = (x + 2)^2 - 1$ increasing?

ANSWER $x \geq -2$

6. Range and boundedness of $y = 4 - x^2$

ANSWER $y \leq 4$, bounded above only

7. Zeros of $f(x) = x^3 - x$

ANSWER $x = 0, \pm 1$

Hard · stretch and reason

7 PROBLEMS

1. Sign intervals of $\frac{x-1}{x+2}$

ANSWER > 0 on $(-\infty, -2) \cup (1, \infty)$

2. Show $f(x) = \frac{1}{x}$ is odd

ANSWER $f(-x) = -\frac{1}{x} = -f(x)$

3. Is $f(x) = \sqrt{x}$ even, odd or neither? Why?

ANSWER Neither — the domain is not symmetric

4. Prove the sum of two odd functions is odd

ANSWER $(f+g)(-x) = -f(x) - g(x) = -(f+g)(x)$

5. Prove the product of an even and an odd function is odd

ANSWER $f(-x)g(-x) = f(x) \cdot (-g(x))$

6. Find all k making $y = x^2 + kx$ increasing on $x \geq 0$

ANSWER $k \geq 0$

7. Show $\frac{1}{1+x^2}$ is bounded, and give both bounds

ANSWER $0 < y \leq 1$

07 Homework

Homework — 6 tasks

Every parity answer must show $f(-x)$ worked out, not just the verdict.

- Zeros and sign intervals of $f(x) = x^2 - 3x - 10$.
- Test $f(x) = x^4 - 3x^2$ and $g(x) = x^5 - x$ for parity.
- State the intervals on which $y = -x^2 + 8x$ increases and decreases.

4. Say whether each of x^3 , $|x|$ and $\frac{1}{x^2}$ is bounded, and how.
5. Give a function that is increasing on $\mathbb{R}$ and odd.
6. Explain why an even function can never be one-to-one on a symmetric domain containing more than one point.